

Machine Learning Course

Uniform stability, Ridge regression and Generalization

1 Context

We consider the following problem.

- $L = \{z_i = (x_i, y_i)\}_{i=1}^m$ where $x_i \in \mathcal{X} \subseteq \mathbb{R}^d$ and $y_i \in [0, B] = \mathcal{Y}$.
- Let $\ell(\theta, z) = (\theta^T x - y)^2 = c(\theta^T x, y)$ be the loss function and $c(\cdot)$ the associated cost function.
- $\hat{\mathcal{R}}(\theta) = \frac{1}{m} \sum_{j=1}^m \ell(\theta, z_j)$, $\hat{\mathcal{R}}^i(\theta) = \frac{1}{m} \sum_{j=1, j \neq i}^m \ell(\theta, z_j)$
- $R_r(\theta) = \frac{1}{m} \sum_{j=1}^m \ell(\theta, z_j) + \lambda \|\theta\|_2^2$
- $R_r^{\setminus i}(\theta) = \frac{1}{m} \sum_{j=1, j \neq i}^m \ell(\theta, z_j) + \lambda \|\theta\|_2^2$
- The ridge regression problem is then defined as:

$$\min_{\theta} R_r(\theta)$$

2 σ -admissibility

We recall the definition.

Definition 1. A loss function ℓ is σ -admissible if the associated cost function c is convex with respect to its first argument and the following conditions holds

$$\forall y_1, y_2 \in D, \forall y' \in \mathcal{Y}, |c(y_1, y') - c(y_2, y')| \leq \sigma |y_1 - y_2|,$$

where $D = \{y : \exists \theta, \exists x, \theta^T x = y\}$ is the domain of the first argument of c .

Show that the loss function of the ridge regression problem is σ -admissible.

For this question, you can assume that $y_i, y' \in \mathcal{Y} = [0, B]$.

3 Result on the minimizers

We now assume this result:

Lemma 1. Let θ be a minimizer of R_r and $\theta^{\setminus i}$ a minimizer of $R_r^{\setminus i}$, knowing that ℓ is σ -admissible, for any $t \in [0, 1]$:

$$\|\theta\|_2^2 - \|\theta + t\Delta\theta\|_2^2 + \|\theta^{\setminus i}\|_2^2 - \|\theta^{\setminus i} - t\Delta\theta\|_2^2 \leq \frac{t\sigma}{\lambda m} |(\Delta\theta)^T x_i|$$

where $\Delta\theta = \theta^{\setminus i} - \theta$

This lemma requires a classic property of any convex function g (see Slides):

$$\forall x, y, \forall t \in [0, 1], g(x + t(y - x)) - g(x) \leq t(g(y) - g(x)).$$

4 Bounding the elements

In this part, we will use the Hölder inequality where $\frac{1}{p} + \frac{1}{q} = 1$ with $p, q > 0$:

$$\sum_i |x_i z_i| \leq \left(\sum_{i=1} |x_i|^p \right)^{1/p} \left(\sum_{i=1} |z_i|^q \right)^{1/q}$$

1. Select a natural value for t in the previous lemma and write the associated inequality.
2. Develop the left hand side of the inequality you just obtained, it should be equal to a term depending only on $\Delta\theta$.
3. Expand now the right hand side of the inequality obtained in item 1 (you need to apply an inequality).
4. Assume now that $\|x\|_2 \leq K$ for all $x \in X$, propose a bound for $\|\Delta\theta\|_2^2$.
5. Now, recall that θ is the optimal minimizer of R_r . Try to give a bound for $\|\theta\|_2^2$. (Hint: if θ is the optimal minimizer of R_r , then for any parameter θ' from $\mathcal{R}^d$, we can define an inequality with $R_r(\theta)$ and $R_r(\theta')$, choose then a simple θ' to derive a bound).
6. Last thing, we need now to show that the loss is bounded. From the elements above, deduce a bound for the loss function of the regression ridge¹ (Hint: you can make use of the previous result of item 6 and hypothesis made in item 4, you can also try to apply the Hölder inequality after adding an absolute value somewhere).

5 Putting things together

You should now be able to find the stability constant and deduce the generalization bound.

1. Find the stability constant, using the previous results you showed previously.
2. Use the theorem allowing one to derive a generalization bound for any algorithm with uniform stability β/n recalled below to provide a generalization bound for ridge regression.

Theorem 1. *Let S be a training sample of size n and $\delta > 0$. For any algorithm L with uniform stability $\frac{\beta}{n}$ with respect to a loss function ℓ bounded by M , with probability $1 - \delta$, we have:*

$$R_{h_S} \leq \hat{R}_{h_S} + \frac{2\beta}{n} + (4\beta + M) \sqrt{\frac{\ln 1/\delta}{2n}}.$$

3. To prove the result in Section 2, you needed to assume that $|\theta^T x| \leq B$ which is not true in general. Can you now update the bound found in Section 2.
4. Can you define a more general formulation of Lemma 1 to generalize the methodology to other models.

¹Actually, assuming that $\theta^T x \in [0, B]$, we can directly state that the bound of the loss by $2B^2$. But we can actually refine this analysis, this would also apply for the σ -admissibility.